

Checkpoint 1 CS465

HENRY ALLISON, SARUNAS JUODAGALVIS, Colorado State University, Fort Collins, USA

1 Introduction

Throughout our existence, we have always been moving. From hunting food to moving to next climates, humans have always had to get from point A to point B. Throughout all of human history it has been from the point of view of our given consciousness, with capabilities such as teleportation only found in fantasy and Sci-fi entertainment. In the last few years, through the use of VR we have been experimenting with other worldly forms of movement, like flight without an airplane or in this case, teleportation. Our aim of this research task was to figure out how people react to locomotion or teleportation in a virtual environment.

We present a VR prototype that compares teleportation and continuous locomotion in a timed gate-based navigation game. Players move through an environment by passing through a sequence of gates while a timer records task performance. The project is designed to investigate which method is more effective for navigating a virtual environment under time pressure.

2 Related Work

Locomotion remains one of the most important design challenges in virtual reality because movement techniques directly affect user performance, comfort, spatial understanding, and overall gameplay experience. In VR applications, teleportation and continuous joystick-based locomotion are among the most common approaches, each offering different tradeoffs. Teleportation is often used to reduce cybersickness and increase comfort, while continuous locomotion can provide more natural and uninterrupted movement through virtual environments.

Langbehn et al. evaluated room-scale VR locomotion techniques by comparing joystick movement, teleportation, and redirected walking. Their study focused on how these techniques influenced navigation and cognitive map formation. The results showed that redirected walking supported the strongest spatial understanding, while joystick-based locomotion was associated with more cybersickness symptoms. Participants also preferred redirected walking and teleportation over joystick locomotion. Although redirected walking is outside the scope of our project, this work is important because it demonstrates the value of comparing locomotion techniques not only by performance, but also by user comfort and spatial outcomes.

Nasiri et al. examined teleportation and joystick-based locomotion over an extended period of use. Their study found that joystick locomotion produced stronger initial navigation performance, while teleportation showed benefits for spatial memory over time. They also observed that some symptoms of cybersickness persisted or increased with repeated exposure for joystick users. This work is relevant to our project because it highlights the tradeoff between short-term performance and other experiential factors in VR navigation. Since our study is focused on a shorter game task rather than a long-term study, their findings suggest that continuous locomotion may offer stronger task performance in our environment from our participants given the length of the game is shorter.

Moore and Lages also examined locomotion techniques but specifically in the context of VR games. Rather than focusing only on navigation efficiency, they examined gameplay-oriented measures such as completion time, level success, and damage taken. Their work is especially relevant to our project because it shows that locomotion should be evaluated within the context of actual game objectives. In our case, players must navigate an environment efficiently

while passing through gates under time pressure, making gameplay performance a meaningful measure alongside general navigation behavior.

Kelly et al. found that rotational self-motion cues improved spatial updating during teleportation in virtual environments, suggesting that turning and orientation design can significantly affect navigation performance in VR. Based on this we debated on whether or not to disable the ability to move using the joysticks when using teleportation, but decided against it as the user can choose whether or not to use that feature very easily.

3 Methodology

Our prototype is a VR navigation game designed to compare locomotion techniques in a controlled but game-like environment. Players are placed in a virtual course containing a sequence of gates that must be reached in order. The system tracks task completion time and gate progress, outputting them to a CSV file. This allows performance to be compared across locomotion conditions.

The current prototype focuses on two locomotion methods: teleportation and continuous locomotion. In the teleportation condition, users point to a destination and instantly move to that location. (However, this capability still needs some ironing with the gate logic, so there is no demo at the moment). In the continuous locomotion condition, users move through the environment using controller input. These conditions were selected because they are among the most common approaches in consumer VR and because prior literature suggests a tradeoff between comfort and movement continuity.

To evaluate the system, participants will complete the same navigation course under both locomotion conditions. Performance measures will include total completion time and missed gates. Subjective measures will include perceived ease of use, comfort, and preference. We also plan to collect brief post-condition feedback regarding workload and discomfort in order to understand how users experience each locomotion technique during play.

4 References

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